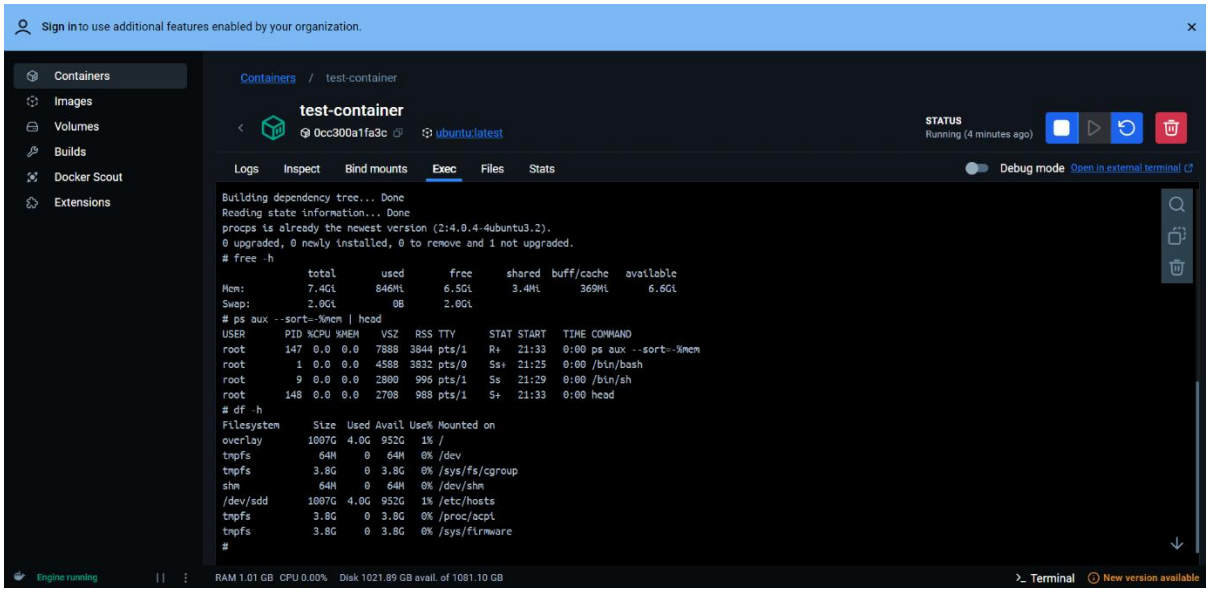
Lab 1

Name:batool wesam

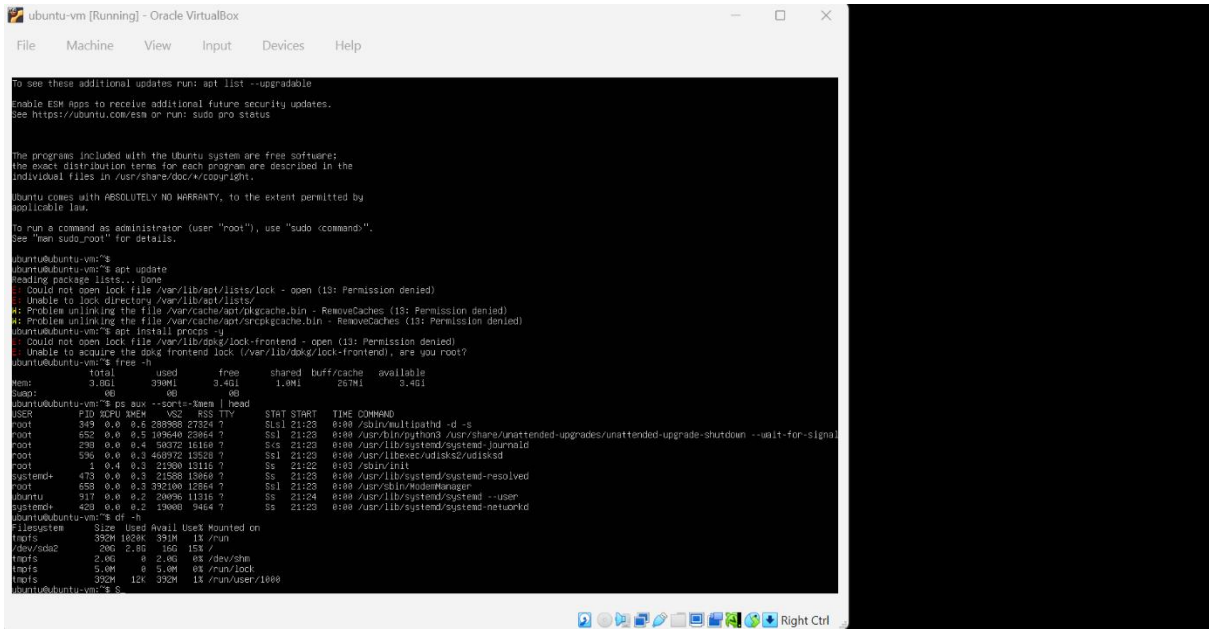
Group:se2

Part A:

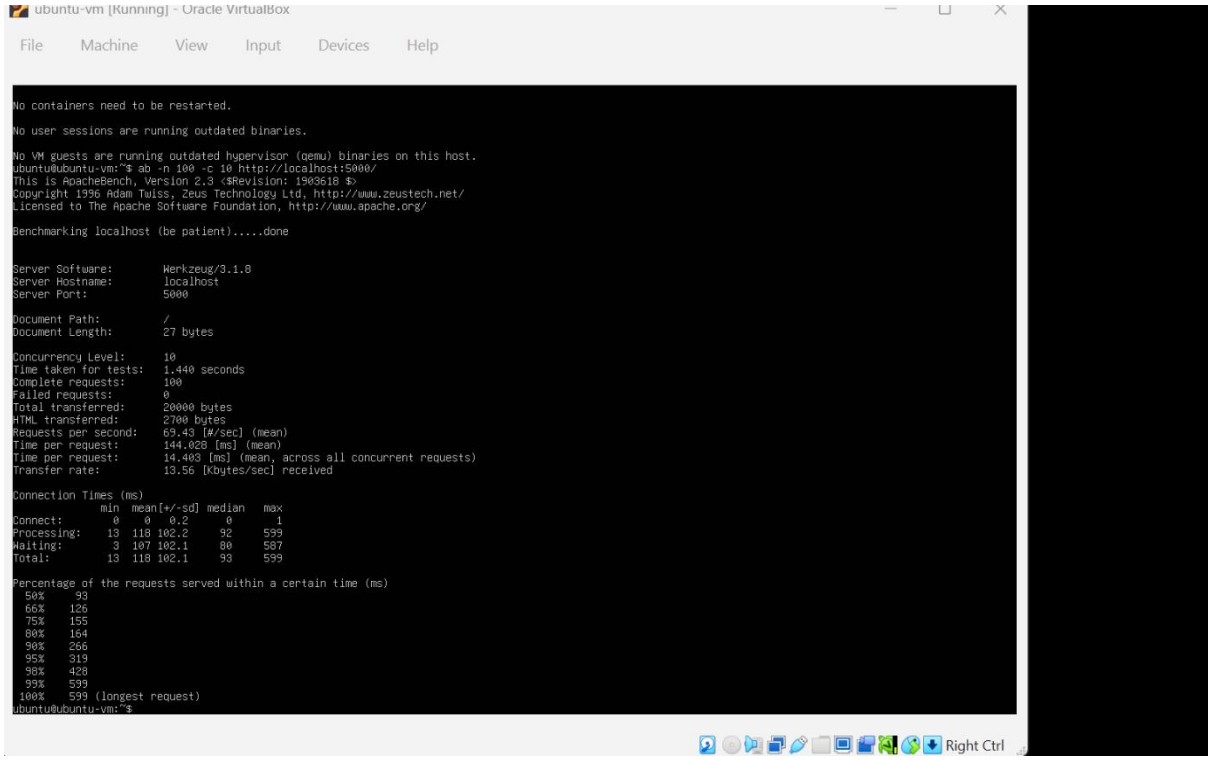
1. Docker container



2-Virtual Machine (VM)



3-VM Setup



Part B – Cloud Shell

1-

```
CLOUD SHELL
Terminal    cloudshell ×                                                    Open Editor
  639 root      20   0    2696   1584   1484 S   0.0   0.0   0:00.00 sleep
  666 batoolw+  10 -10    7280   2696   1944 S   0.0   0.0   0:00.09 tmux
  667 batoolw+  10 -10    8796   5780   3924 S   0.0   0.1   0:00.06 bash
 1228 root      10 -10   13996   9868   8488 S   0.0   0.1   0:00.01 sshd
 1230 root      10 -10   13996   9780   8404 S   0.0   0.1   0:00.01 sshd
 1232 batoolw+  10 -10   14256   6456   4796 S   0.0   0.1   0:00.00 sshd
 1233 batoolw+  10 -10   14256   6440   4784 S   0.0   0.1   0:00.01 sshd
 1234 batoolw+  10 -10    7340   3692   3444 S   0.0   0.0   0:00.00 bash
 1237 batoolw+  10 -10    7604   4472   3868 S   0.0   0.1   0:00.00 bash
 1240 batoolw+  10 -10    7340   3820   3572 S   0.0   0.0   0:00.00 bash
 1241 batoolw+  10 -10    7340   3640   3380 S   0.0   0.0   0:00.00 start-shell.sh
 1242 batoolw+  10 -10    6880   3452   3148 S   0.0   0.0   0:00.00 tmux
 1243 batoolw+  10 -10    8796   5800   3944 S   0.0   0.1   0:00.04 bash
```

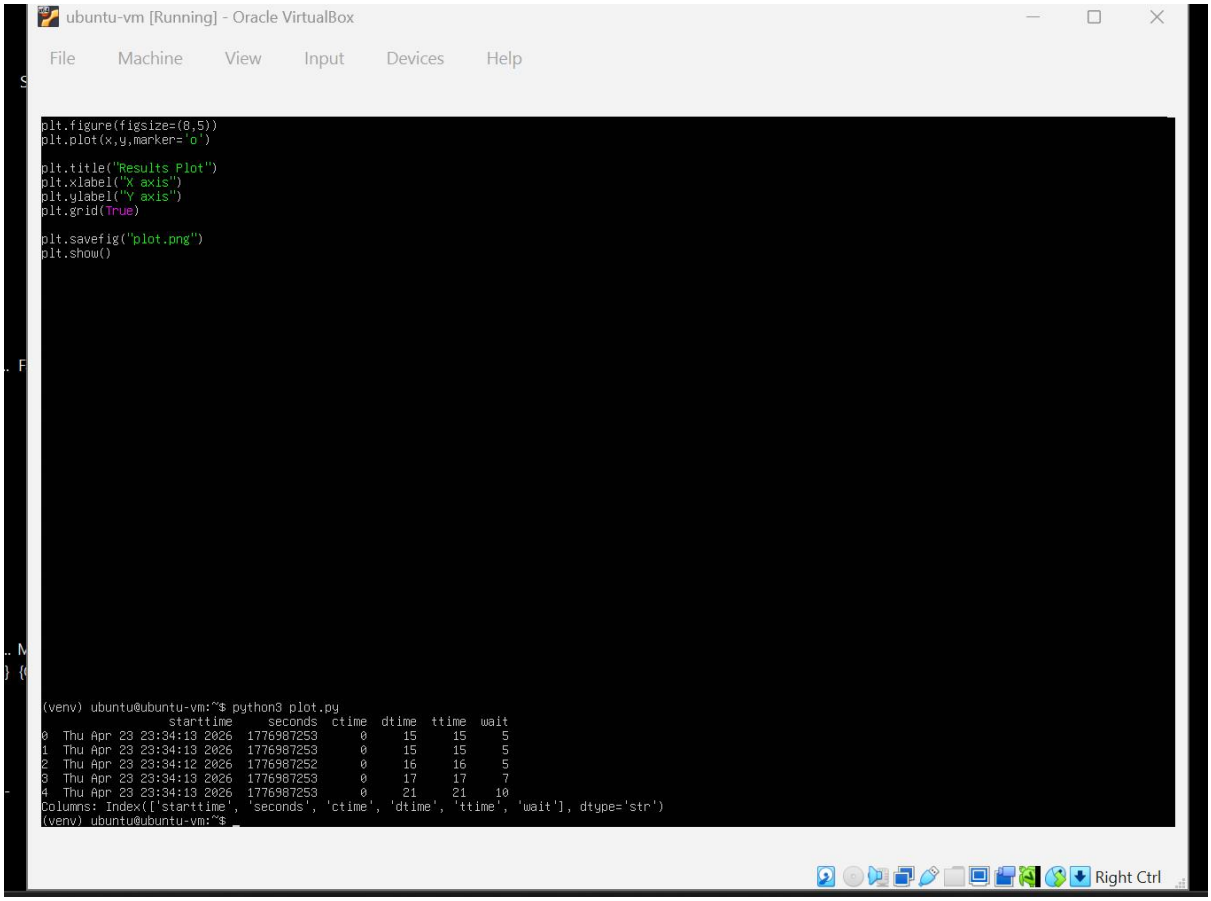
2-

```
Cloud Shell Editor
cloudshell ×
Gemini CLI is available in Cloud Shell terminal! Type gemini to try it. Learn more        Don't show again  Dismiss
   68 root      10 -10   12024   4012   2900 S   0.0   0.0   0:00.02 sshd
  312 root      20   0 2087572  84384  61172 S   0.0   1.0   0:00.78 dockerd
  415 root      20   0 1232860   8188   7056 S   0.0   0.1   0:00.01 editor-proxy
  416 root      20   0   16328   6460   5532 S   0.0   0.1   0:00.01 sudo
  426 root      20   0 1226100   2660   1976 S   0.0   0.0   0:00.00 tmux-agent
  455 root      20   0 1719468  45892  32572 S   0.0   0.6   0:04.67 containerd
  639 root      20   0    2696   1584   1484 S   0.0   0.0   0:00.00 sleep
  666 batoolw+  10 -10    7280   2696   1944 S   0.0   0.0   0:00.09 tmux
  667 batoolw+  10 -10    8796   5780   3924 S   0.0   0.1   0:00.06 bash
 1228 root      10 -10   13996   9868   8488 S   0.0   0.1   0:00.01 sshd
 1230 root      10 -10   13996   9780   8404 S   0.0   0.1   0:00.01 sshd
 1232 batoolw+  10 -10   14256   6456   4796 S   0.0   0.1   0:00.00 sshd
 1233 batoolw+  10 -10   14256   6440   4784 S   0.0   0.1   0:00.01 sshd
 1234 batoolw+  10 -10    7340   3692   3444 S   0.0   0.0   0:00.00 bash
 1237 batoolw+  10 -10    7604   4472   3868 S   0.0   0.1   0:00.00 bash
 1240 batoolw+  10 -10    7340   3820   3572 S   0.0   0.0   0:00.00 bash
 1241 batoolw+  10 -10    7340   3640   3380 S   0.0   0.0   0:00.00 start-shell.sh
 1242 batoolw+  10 -10    6880   3452   3148 S   0.0   0.0   0:00.00 tmux
 1243 batoolw+  10 -10    8796   5800   3944 S   0.0   0.1   0:00.04 bash
 1649 batoolw+  10 -10   11892   5460   3336 R   0.0   0.1   0:00.00 top
batoolwesam26@cloudshell:~$ df -h
Filesystem                         Size  Used Avail Use% Mounted on
overlay                             95G   50G   45G  53% /
tmpfs                               64M     0   64M   0% /dev
/dev/sda1                           95G   50G   45G  53% /root
/dev/disk/by-id/google-home-part1  4.8G  1.4M  4.6G   1% /home
/dev/root                          2.0G  1.3G  715M  64% /usr/lib/modules
shm                                 64M     0   64M   0% /dev/shm
batoolwesam26@cloudshell:~$
```

Part C – Benchmarking

```
File    Machine    View    Input    Devices    Help

venv) ubuntu@ubuntu-vm:~$ head results.tsv
tarttime        seconds ctime   dtime   ttime   wait
u Apr 23 23:34:13 2026          1776987253      0       15      15      5
u Apr 23 23:34:13 2026          1776987253      0       15      15      5
u Apr 23 23:34:12 2026          1776987252      0       16      16      5
u Apr 23 23:34:13 2026          1776987253      0       17      17      7
u Apr 23 23:34:13 2026          1776987253      0       21      21      10
u Apr 23 23:34:12 2026          1776987252      1       21      22      10
u Apr 23 23:34:12 2026          1776987252      1       21      22      10
u Apr 23 23:34:12 2026          1776987252      0       23      23      10
u Apr 23 23:34:13 2026          1776987253      0       23      23      12
venv) ubuntu@ubuntu-vm:~$ python3
ython 3.12.3 (main, Mar 23 2026, 19:04:32) [GCC 13.3.0] on linux
ype "help", "copyright", "credits" or "license" for more information.
>> python3 plot.py
 File "<stdin>", line 1
   python3 plot.py
         ^^^^
yntaxError: invalid syntax
>> python3 plot.py
 File "<stdin>", line 1
   python3 plot.py
         ^^^^
yntaxError: invalid syntax
>> nano plot.py
 File "<stdin>", line 1
   nano plot.py
        ^^^^
```

Part C – Data File + Python Errors



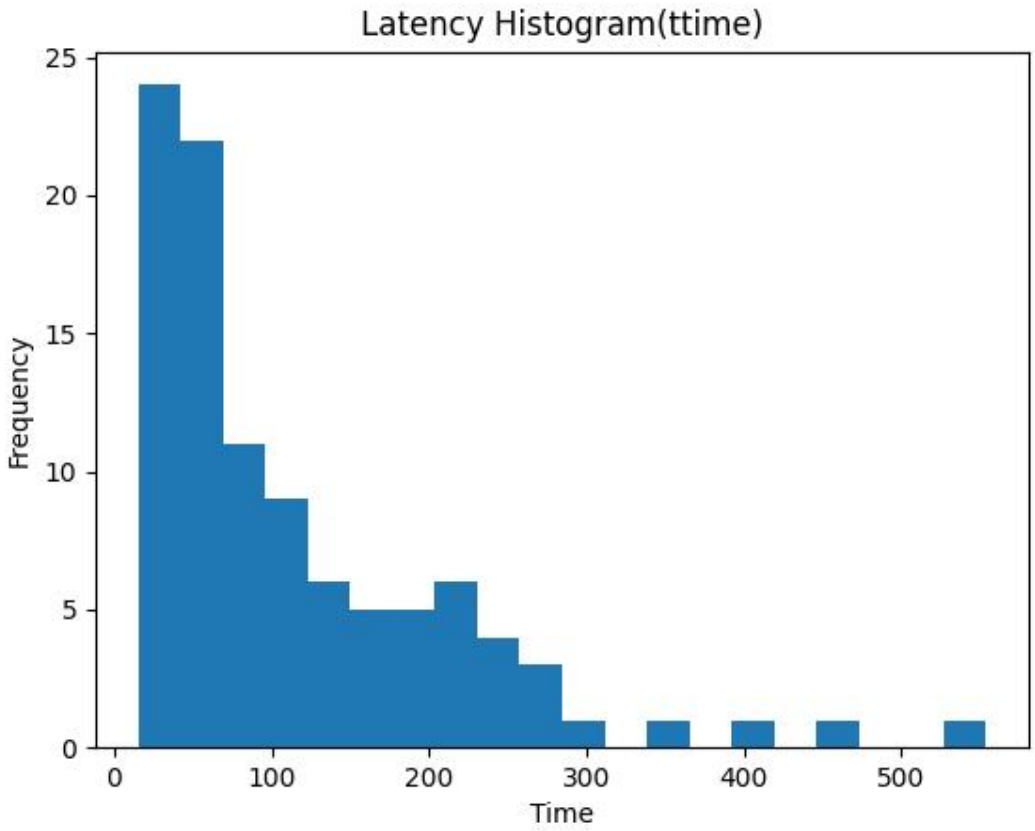
Part C – Plot Script

ubuntu-vm [Running] - Oracle VirtualBox

File Machine View Input Devices Help

```
unning kernel seems to be up-to-date.

o services need to be restarted.

o containers need to be restarted.

o user sessions are running outdated binaries.

o VM guests are running outdated hypervisor (qemu) binaries on this host.
venv) ubuntu@ubuntu-vm:~$ sudo mount /dev/cdrom /mnt
ount: /mnt: WARNING: source write-protected, mounted read-only.
venv) ubuntu@ubuntu-vm:~$ sudo /mnt/VBoxLinuxAdditions.run
erifying archive integrity...  100%   MD5 checksums are OK. All good.
ncompressing VirtualBox 7.2.8 Guest Additions for Linux  100%
irtualBox Guest Additions installer
irtualBox Guest Additions: Setting up modules
irtualBox Guest Additions: Building the VirtualBox Guest Additions kernel
odules.  This may take a while.
irtualBox Guest Additions: To build modules for other installed kernels, run
irtualBox Guest Additions:   /sbin/rcvboxadd quicksetup <version>
irtualBox Guest Additions: or
irtualBox Guest Additions:   /sbin/rcvboxadd quicksetup all
irtualBox Guest Additions: Building the Guest Additions 7.2.8 modules for
ernel 6.8.0-110-generic.
irtualBox Guest Additions: Building the main module.
irtualBox Guest Additions: Building the shared folder support module.
irtualBox Guest Additions: Building the graphics driver module.
irtualBox Guest Additions: Updating initramfs
pdate-initramfs: Generating /boot/initrd.img-6.8.0-110-generic
irtualBox Guest Additions: Running kernel modules will not be replaced until
he system is restarted or 'rcvboxadd reload' triggered
irtualBox Guest Additions: Could not find the X.Org or XFree86 Window System,
kipping.
irtualBox Guest Additions: Starting.
irtualBox Guest Additions: reloading kernel modules and services
irtualBox Guest Additions: kernel modules and services 7.2.8 r173730 reloaded
irtualBox Guest Additions: NOTE: you may still consider to re-login if some
ser session specific services (Shared Clipboard, Drag and Drop, Seamless or
uest Screen Resize) were not restarted automatically
venv) ubuntu@ubuntu-vm:~$ ls /media
f_histogram
venv) ubuntu@ubuntu-vm:~$ ls
op.py  hist.png  plot.py  plot.py  results.tsv  venv
venv) ubuntu@ubuntu-vm:~$ cp hist.png /media/sf_histogram/
p: cannot stat '/media/sf_histogram/hist.png': Permission denied
venv) ubuntu@ubuntu-vm:~$ sudo cp hist.png /media/sf_histogram/
venv) ubuntu@ubuntu-vm:~$ ls /media/sf_histogram
s: cannot open directory '/media/sf_histogram': Permission denied
```

Latency Histogram



1-AWS's Nitro system offloads traditional hypervisor responsibilities (networking, storage, security) into dedicated hardware cards instead of keeping everything in software.

2-Use VMs when Strong security boundaries are required,Different OS requirements,Legacy or monolithic applications that depend on full OS features ,Untrusted workloads (VMs isolate better than containers)

Use containers when want fast startup and scaling or building microservices